

PROFILE

Electronics and Communication Engineering Final-year ECE student focused on Embedded Systems, Embedded Software, IoT, Robotics, and AI-driven automation. Skilled in C, Embedded C, microcontrollers, ESP32, STM32, sensors, and hardware–software integration. Experienced in developing practical projects involving autonomous systems, LiDAR, automation, and intelligent control. Passionate about building reliable embedded solutions for real-world applications.

EDUCATION

KUMARAGURU COLLEGE OF TECHNOLOGY (2022-2027)

- Bachelor of Electronics And Communication Engineering

BHARATHI VIDYA BAVAN MATRIC HR SEC SCHOOL (2019-2022)

- Course - Computer Science
- Score: 83.1%

CONTACT

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SACHIN J

Embedded Systems & IoT Engineer | ECE Student

PROJECTS AND RESEARCH

Autonomous Outdoor Path Cleaning Robot

Raspberry Pi | LiDAR | Robotics | IoT

- Designed an autonomous cleaning robot intended for systematic outdoor path cleaning.
- Incorporated LiDAR-based obstacle detection to support navigation and environmental awareness.
- Focused on integrating navigation, monitoring, motor control, and cleaning operations into a unified robotic system.
- Explored solar-assisted power to support energy-efficient operation.

Research Focus: Autonomous navigation, LiDAR sensing, robotic automation, and sustainable power.

FixNear – Local Services & Rental Marketplace

Web Development | UI/UX | Location Services | Marketplace

- Designed and developed a local services and rental marketplace connecting customers with nearby service providers and rental owners.
- Structured the platform around local services, vehicle rentals, home appliance rentals, and acting driver services.
- Designed location-based discovery, category filtering, budget selection, and rental-duration options to improve service discovery.
- Planned provider profiles and customer workflows for contact, communication, booking, and rental management.
- Focused on creating an accessible and scalable marketplace experience with smart-search and multilingual-support concepts.

Research Focus: Location-based marketplaces, digital service platforms, user experience, and rental management.

Android Call Assistant

Kotlin | Android Studio | Voice Interaction

- Designed an Android-based call assistant to support intelligent handling of incoming calls.
- Developed workflows for call acceptance and rejection based on user commands and availability.
- Explored voice-based interaction and automated responses for situations in which the user could not answer.
- Focused on making mobile communication more convenient through hands-free interaction and automation.

Research Focus: Android development, voice interaction, mobile automation, and context-aware assistants.

ON-CAMPUS PROJECTS

MICROCOSM

Solar-Powered Mobile Charging System

Solar Energy | Power Management

Designed a portable solar charging system with rechargeable battery storage and power-management concepts.
Technical Focus: Renewable energy, battery storage, power management, and portable electronics.

FORGE

Car Accident Safety Monitor

ESP32 | Sensors | Automotive Safety | IoT

Designed a sensor-based accident detection and emergency-alert system with location-sharing concepts..
Technical Focus: Embedded sensor systems, crash detection, automotive safety, and IoT-based emergency alerts.

DRONE TECHNOLOGY

Multi-Functional Autonomous Drone for Emergency & Environmental Applications

Pixhawk | Raspberry Pi | ESP32 | GPS | 4G/LTE

Build an autonomous drone for environmental monitoring, surveillance, and emergency delivery.
Technical Focus: Autonomous drones, environmental sensing, remote communication, and emergency response.

CERTIFICATIONS

TECHNOLOGY INTERESTS

- Embedded Software Engineering
- Microcontroller Programming
- IoT and Sensor-Based Systems
- Autonomous Robotics
- AI-Driven Automation
- Automotive Embedded Systems
- Hardware–Software Integration
- Real-Time Embedded Applications

PROFESSIONAL SKILLS

- Problem Solving
- Analytical Thinking
- Teamwork and Collaboration
- Communication
- Adaptability
- Quick Learning

LANGUAGES

- **English:** Fluent
- **Tamil:** Fluent
- **German:** Basic